EX NO 3

LEARNING AND ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTERS

AIM:

To understand the IP addressing scheme and manually add IP addresses to computer on network.

THEORY:

1) An IP (Internet Protocol) address is an unique numerical label assigned to each device connected to a computer network.

2) IPv4 addresses are 32-bit numbers written in dotted-decimal notation.

3) IPV6 uses 128-bit address.

4) Subnet mask determines the network and host portions.

5) Manual (static) IP assignment is used for server and network devices requiring a permanent address.

6) DHCP assigns IP dynamically.

PROCEDURE:

1) In windows: Open Network and sharing center → change Adapter settings → Right click Local Area connection → Properties → IPv4 → Set IP manually.

2) Enter IP Address: 192.168.1.10, Subnet Mask: 255.255.255.0, Default gateway: 192.168.1.1.

3) Enter DNS Server: 8.8.8.8 (Google DNS), click OK.

4) In linux : Use command 'Sudo ip addr add 192.168.1.10/24 dev eth0'.

5) To set gateway in linux 'sudo ip route add default via 192.168.1.1'

6) For persistent configuration in Linux, edit / etc / network / interfaces or use nmcli.

7) Verify the configuration using 'ipconfig' or 'ip a'

8) Ping the default gateway to confirm network connectivity.

9) Ping another computer on the same subnet to verify communication.

CODE:

```
sudo ip addr add 192.168.1.10/24 dev eth0
sudo ip route add default via 192.168.1.1
ip a show eth0
Ping 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.addresses 192.168.1.10/24
nmcli con mod 'Wired connection 1' ipv4.gateway 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.method manual
nmcli con up 'Wired connection 1'
```

RESULT:

Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.